



Collaborative robotics

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A Voice-Controlled Fully Assisted Elbow Exoskeleton for Vocal Trivial Pursuit

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Abstract and graphical abstract

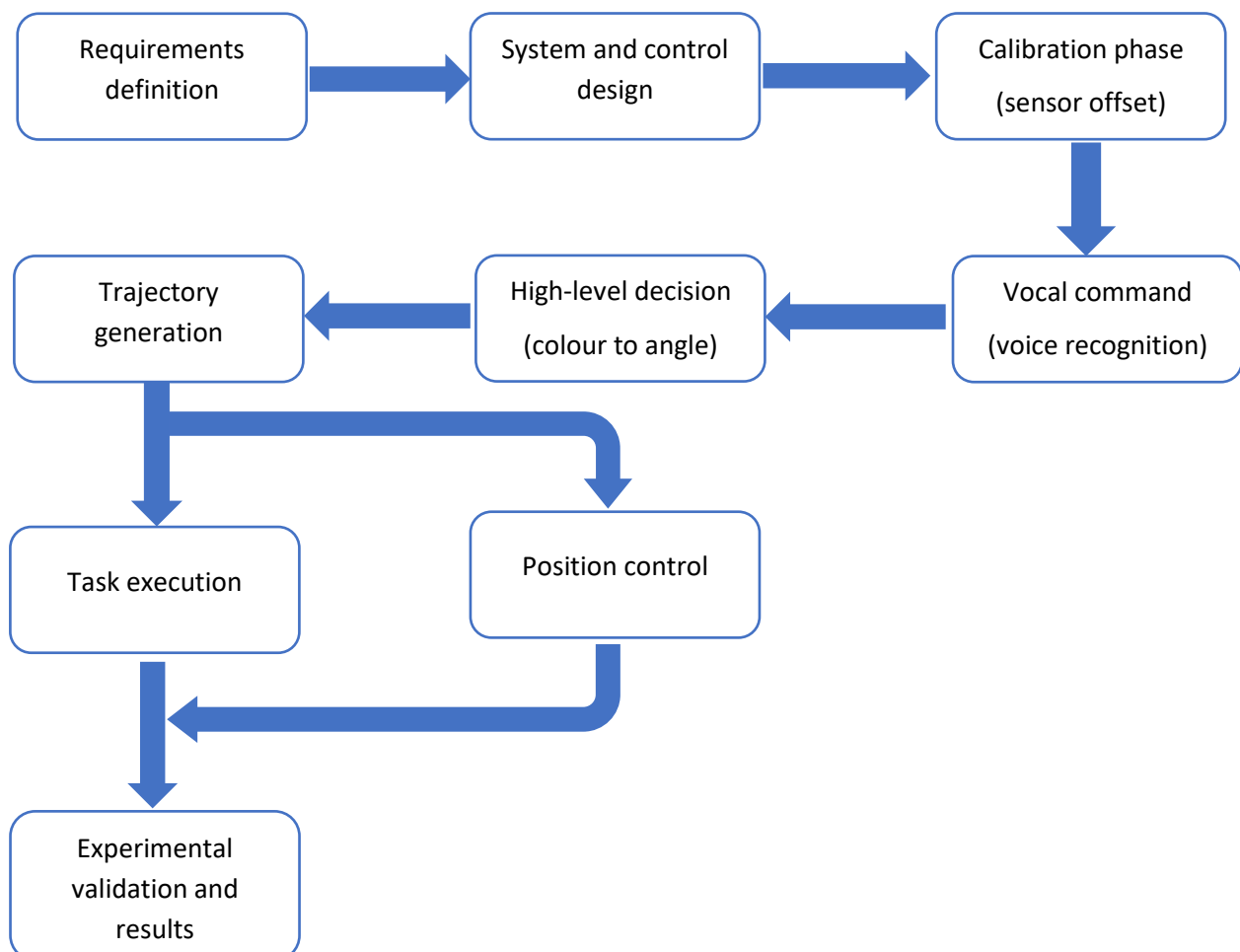
This project presents *Vocal Trivial Pursuit*, an assistive robotic system designed to meet the requirements of users with complete loss of voluntary upper-limb function, such as subjects with spinal cord injury, ensuring safe and **fully assisted elbow movements** toward discrete target positions.

The system is based on a powered elbow exoskeleton and enables active participation in a **quiz-like task** by selecting answers through **voice commands**, which are mapped to predefined elbow configurations.

User intention is detected exclusively via **speech recognition**. Each recognized vocal command corresponds to a colour, and each colour is associated with a specific **target elbow angle** (0° , 45° or 90°). Once a valid command is identified, the exoskeleton autonomously moves the user's elbow to the selected target through position control, which also acts as a **continuous safety mechanism** by monitoring tracking errors and preventing unsafe behaviours.

The system operates wirelessly during normal use, with **no physical connection to external devices**. A wired USB connection is required only during the calibration phase, during which sensor offsets and the mapping between encoder readings and joint angles are identified. After calibration, the exoskeleton functions fully standalone.

Assisted movements are executed following a **minimum-jerk trajectory**, allowing comparison between simple and human-like motion generation strategies while maintaining predictability and safety during human-robot interaction. Experimental tests demonstrate the **correct execution** of vocal commands, reliable target reaching, and safe task achievement in accordance with the defined system requirements.



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1. Motivation of the project

Subjects with complete spinal cord injury often present severe motor impairment of the upper limb, losing the ability to perform basic activities of daily living despite preserved cognitive and communicative functions. This loss of independence has a strong impact on participation, social life, and perceived quality of life, and motivates the development of assistive robotic devices able to restore functional actions that the user cannot perform autonomously.

Within this context, upper-limb exoskeletons can provide an orthotic effect, i.e. an immediate improvement in functional performance while the device is worn, by compensating for the absence of voluntary elbow torque generation. For individuals with no residual strength at the elbow, it is crucial that the system guarantees safe, predictable, and repeatable movements, since the limb is fully driven by the robot.

Moreover, these users typically maintain intact cognitive abilities and speech, which can be exploited to implement intuitive, hands-free intention detection strategies based on vocal commands. Combining voice control with an elbow exoskeleton allows users to actively select targets and initiate movements, preserving a sense of agency even in the absence of voluntary motor control.

Comfort is a central requirement for wearable collaborative robots used in healthcare, where physical human–robot interaction occurs in close contact and for prolonged periods. Smooth, human-like motion trajectories, controlled joint range of motion, and robust monitoring of abnormal conditions are required to minimize discomfort and reduce the risk of unsafe transients or unexpected behaviours.

Finally, to support real-life assistive scenarios beyond constrained laboratory environments, the system should operate in a portable and wireless configuration, allowing the exoskeleton to be used at the bedside or at home without cumbersome cabling and with minimal setup burden for caregivers and users.

2. Objective of the work

The objective of this work is to design, develop, and validate an assistive robotic system for the *Vocal Trivial Pursuit* task, based on a 1-degree-of-freedom powered elbow exoskeleton. The system is intended for subjects with severe motor impairments of the upper limb, such as patients with spinal cord injury, characterized by the absence of voluntary elbow flexion–extension (Muscle Research Council scale equal to 0 for both biceps and triceps), but preserved cognitive and speech abilities.

Specifically, the work aims to: define the kinematic and dynamic model of the human–exoskeleton system as a preliminary step to inform the control design. Based on this model, a position-controlled assistive strategy was developed to provide full motor assistance, ensure accurate target reaching, and guarantee continuous safety monitoring during human–robot interaction.

User intention is detected exclusively through vocal commands, which are mapped to predefined elbow target angles associated with the possible answers in a quiz-like scenario. During task execution, the system operates wirelessly, allowing the exoskeleton to be controlled without a physical connection. Motion execution is implemented using minimum-jerk trajectories, to ensure smooth, predictable, and safe arm movements.

Experimental tests are proposed to validate the system on multiple participants assessing target-reaching accuracy, latency, robustness, repeatability, and usability with respect to the predefined functional and non-functional requirements.

3. Requirements

3.1 Functional requirements

Requirement	Description	Means of verification
<i>Voice command set</i>	The system shall accept a fixed set of vocal commands and reject unrecognized input.	Command trials with valid/invalid inputs; only predefined commands trigger actions.
<i>Command-to-target mapping</i>	Each valid vocal command shall be uniquely mapped to a predefined elbow target position.	Repeated commands; verify one-to-one command–target association.
<i>Range of motion</i>	The system shall execute elbow flexion to three discrete target angles.	Verify that commanded motions terminate at one of the predefined target angles.
<i>Motion execution</i>	Upon a valid command, the system shall autonomously move the elbow to the target using position control.	Verify automatic trajectory execution after command recognition.
<i>Target reaching accuracy</i>	Final elbow position error shall be lower than 5°.	Encoder-based steady-state error < 5° over repeated trials.
<i>Stuck condition safety</i>	If stationary for ten samples outside equilibrium, actuation shall stop.	Induce blockage; verify stop after ten samples.
<i>Sensor-based position feedback</i>	Elbow position shall be measured and used for control and safety.	Verify sensor feedback used in control loop and safety logic.

3.2 Non-functional requirements

Requirement	Description	Means of verification
<i>System latency</i>	Command-to-motion latency shall be < 5 s.	Measure time from command end to motion onset.
<i>Safety on invalid input or disconnection</i>	Invalid input or Wi-Fi loss shall result in a safe, non-moving state.	Inject invalid commands and disconnect Wi-Fi; verify no motion.
<i>Session reliability</i>	Continuous operation for at least 50 min without failure.	Continuous session test > 50 min.
<i>Usability</i>	Trade-off between wearing comfort and ease of use	Patient assessment with survey about quality of the system and comfortability
<i>Recoverability</i>	After failures, normal operation shall be restored quickly.	Disconnect/reconnect Wi-Fi; verify recovery < 2 min.
<i>Multi-subject validation</i>	System robustness shall be assessed across different users.	Repeat protocol on multiple users and compare outcomes.

4. Materials and methods

4.1 Hardware

The hardware used in this project is the EduExo Pro kit [1], employed as an assistive device for subjects with spinal cord injury with no residual upper-limb function (Muscle Research Council, MRC = 0). The EduExo Pro is a wearable, portable exoskeleton designed to actively assist the elbow joint and passively support the shoulder. Although originally designed for the left arm, it can be easily adapted for right-arm use.

The kinematic architecture matches human joints, implementing a 1 DoF active rotational joint for elbow flexion-extension and a passive 2 DoF ball joint at the shoulder. The actuation strategy is based on a servomotor at the elbow, which gives the necessary torque to move the forearm. It is helped by a passive torsion spring at the shoulder, which is used to stabilize the movement by applying gravitational compensation. During our project we use the hardware to move the forearm in a range between 0° and 90°, which is sufficient to complete our task.

The low-level control strategy of the EduExo Pro is carried out by the microcontroller Arduino Nano 33 IoT. The main components and connections involved in our project are:

- Actuator: a servomotor (which can be commanded only in position mode) with an internal angle sensor, which is connected to an analog port for the position control feedback.
- Sensors: a force sensor in the forearm structure, which is used to measure the interaction of forces. There is also an IMU in the Arduino Nano 33 IoT, which can be used to measure shoulder movements.
- Supply system: the energy to the system is supplied by a battery pack of six AA type batteries to ensure portability.
- Computer: It is needed for the intention detection strategy, which is based on vocal command detection. The PC is used to elaborate the vocal signals of the user and translate them into control commands, which are sent to the microcontroller by a Wi-Fi connection. Thus, we can talk about passive trigger actuation.
- Microphone: It assumes the role of coupling between the user and the hardware. We can both use the computer integrated microphone or an external microphone which must be plugged in with the PC.

4.2 Kinematic and dynamic model

Kinematic Model

Although the EduExo Pro device and the human upper limb can be described as a 2-DoF system, the model is intentionally simplified for the Vocal Trivial Pursuit scenario. The subject is seated with the upper arm aligned with the torso, shoulder motion is neglected, and no voluntary torque is generated due to full paralysis. The system is therefore modelled as a 1-DoF elbow flexion mechanism, with the elbow angle $\theta(t)$ as the only generalized coordinate. The corresponding kinematic model is shown in Fig. 1.

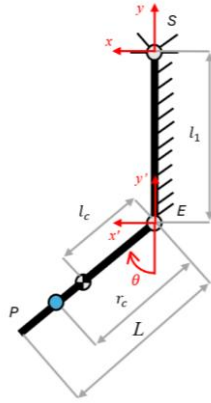


Figure 1: Kinematic model

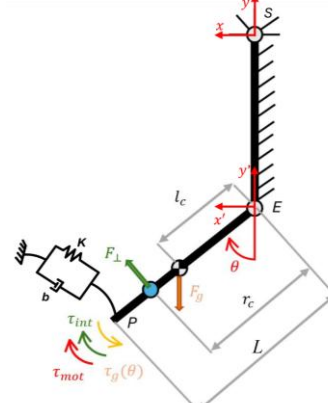


Figure 2: Dynamic model

The forward kinematics of the wrist and the corresponding differential kinematics are defined as:

$$p_P(\theta) = \begin{bmatrix} x_P(\theta) \\ y_P(\theta) \end{bmatrix} = \begin{bmatrix} L \sin \theta \\ -L \cos \theta \end{bmatrix} \Rightarrow \dot{p}_P(\theta) = \frac{\partial p_P}{\partial \theta} \dot{\theta} = \begin{bmatrix} L \cos \theta \\ L \sin \theta \end{bmatrix} \dot{\theta}$$

Dynamic model

The human forearm and hand are modelled as an equivalent rigid segment. According to anthropometric tables [2], the combined **forearm-hand segment** is defined from the elbow axis to the ulnar styloid and is characterized by a mass fraction $m = 0.022 M$, COM location $l_c = 0.682 L$ from the proximal end. For a male subject with $H = 1.80 m$ and $M = 70 kg$, these parameters are used to compute the equivalent mass m , COM distance l_c , and elbow inertia J . Based on these assumptions, the system dynamics describes the rotation of the equivalent forearm-hand segment about the elbow joint as shown in Fig. 2. The model is derived in joint space and includes gravitational loading, passive viscoelastic effects, and interaction torques measured at the forearm. Since the subject is fully paralyzed, the **active human torque is assumed null**: $\tau_{hum}(t) = 0$.

- $\tau_{mot}(t)$: equivalent actuator torque applied by the device at the elbow joint.
- $\tau_g(\theta)$: gravitational torque of the moving segment equal to $m g \cdot l_c \sin \theta$
- $\tau_{int}(t) = r_c F_{\perp}(t)$: interaction torque generated by external forces measured by the force sensor.
- $b \dot{\theta}$: equivalent viscous damping term modelling dissipative joint effects.
- $k(\theta - \theta_0)$: equivalent linear stiffness term modelling passive elastic effects around θ_0 .

The complete equation of motion is: $J \ddot{\theta} + b \dot{\theta} + k(\theta - \theta_0) + m g \cdot l_c \sin \theta = \tau_{mot}(t) + \tau_{int}(t)$

This formulation highlights that, in the absence of voluntary torque, the actuator must overcome **gravity**, **passive viscoelastic effects**, and any **external interaction torque** measured along the forearm.

4.3 Control design

Hardware level of compliance

Within the *Vocal Trivial Pursuit* project, hardware-level compliance refers to the selection of a mechatronic architecture (actuation and sensing) that enables safe and tuneable physical interaction, while the device provides full assistance toward predefined targets selected via voice commands.

The **EduExo Pro** platform adopts the "Geared motor + external Force/Torque sensor" approach: a position-controlled actuator (a servo motor with an internal controller and geared transmission) is paired with an interaction force sensor mounted at the forearm interface. This configuration does not possess intrinsic mechanical compliance characterized by fixed physical stiffness; rather, it enables **active compliance**.

The measured interaction force serves as the input for an **admittance controller**, which generates a position reference (or offset) to be sent to the joint internal position controller. This allows the robot to track the target trajectory while simultaneously yielding in a controlled manner, limiting interaction forces in the presence of contacts, resistance, or abnormal conditions.

Low-level control solutions for compliance

Impedance control

Explicit impedance control aims to shape the robot's dynamics in terms of equivalent stiffness and damping by commanding the actuator to deliver a torque proportional to position and velocity errors.

In **EduExo Pro**, however, the interface accessible via Arduino only allows sending a position setpoint through a PWM signal, while the position tracking is handled by the servo internal controller. Consequently, an inner torque loop is not accessible at the application level. Therefore, a classical torque-based impedance implementation cannot be directly realized.

Admittance control

Admittance control is the conceptual dual of impedance: instead of generating an output force/torque based on a position error, it utilizes a measured force as input to compute a motion command (position or velocity). It is often described as *position-based impedance* or *impedance with force feedback*, since compliance is achieved by modifying the position reference in real-time as a function of the measured interaction force.

The control scheme consists of three main components, as shown in Fig. 3:

- **The servo motor:** Provides a robust, high-performance inner position loop responsible for tracking the setpoint.
- **The force sensor:** Measures the interaction between the exoskeleton and the user, providing the physical signal required to close the outer loop.
- **The outer loop:** Computes the necessary modification to the position of reference to achieve compliant and safe behaviour.

Functional implementation

Motion reference generation

At a high level, the voice command selects a target (among three options), defining a time-dependent reference trajectory for the elbow, denoted as $\theta_{ref}(t)$.

Interaction measurement (force sensor)

During execution, the force sensor produces an analog signal acquired via the analogRead function. The hardware requires calibration to correctly represent forces in both directions (flexion/extension). This step is critical: without proper calibration, the outer loop cannot correctly interpret the interaction, rendering the compliance behaviour unpredictable.

Admittance filter (force $\rightarrow$ motion)

The calibrated force signal is fed into an admittance filter $A(s)$, which represents the virtual dynamics mapping force into a motion contribution.

$$A(s) = (K_p + sK_d)^{-1}$$

The filter generates a position correction $\theta_{adm}(t)$. As the contact force increases (due to contact, obstacles, or spasm), the controller produces an offset that reduces the exchanged force, effectively causing the system to yield.

Final command to the servo

The final command sent to the servo is defined as

$$\theta_{cmd}(t) = \theta_{adm}(t) + \theta_{ref}(t)$$

The system does not command torque directly; instead, it modifies the reference trajectory in real time based on the measured force. This generates a **virtual compliance** that ensures safe and smooth interaction.

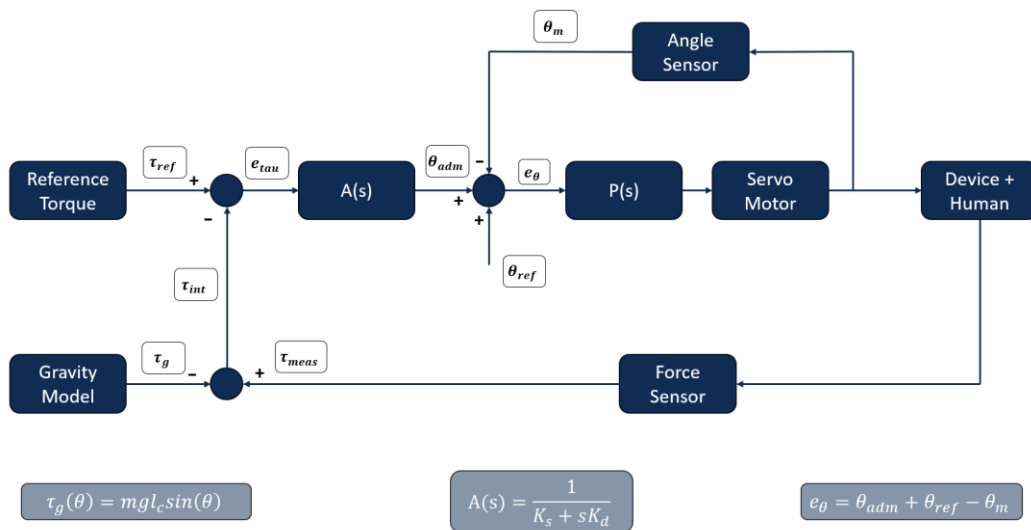


Figure 3: Admittance control block diagram

High-Level Control Solution (Vocal Trivial Pursuit)

The high-level control is not strictly constrained by the specific low-level hardware interface but governs the overall human–robot interaction.

Its main functions include:

- Target selection based on voice commands.
- Generation of the motion reference toward the selected target.
- On-demand activation of assistance through a cognitive trigger (voice command)

Taxonomically, this solution corresponds to a **Passive, Triggered Trajectory-Following Mode** [3]. The robot executes a predefined trajectory in a passive mode (characterized by stiff behaviour and high corrective force) only when the vocal trigger is detected. The high-level layer defines the task logic (target selection, trajectory generation, and trigger-based activation), whereas the compliance required for safety is delegated to the low-level controller.

Motion planning: Minimum-Jerk trajectories

The assisted elbow motion follows **minimum-jerk trajectories**. Based on the biological principle that the human body tends to minimize energy consumption, minimizing jerk reduces motion abruptness. This results in smooth profiles with continuity up to acceleration, yielding more natural and safer movements in assistive human–robot interaction.

“The human body always tries to minimize energy consumption”

Mathematically, minimizing jerk implies a trajectory whose sixth derivative is zero, leading to a fifth-order polynomial trajectory profile. The polynomial coefficients are obtained by imposing rest-to-rest boundary conditions (initial/final position, with zero velocity and acceleration at the endpoints).

Imposing **rest-to-rest boundary conditions** guarantees that the motion starts and ends without residual velocity or acceleration, preventing impulsive transients and overshooting at the target. When combined with a **minimum-jerk** criterion, the resulting quintic trajectory enforces continuity up to acceleration and limits abrupt changes in actuation. This is particularly important in assistive human–robot interaction, where reducing motion discontinuities improves comfort and lowers the risk of unsafe interaction peaks.

Intention Detection Strategies (Trigger mode – Vocal Control)

The choice of a vocal Intention Detection Strategy (IDS) in trigger mode involves a trade-off: maximizing usability and adaptability to the user’s residual capabilities, while balancing performance (reliability and robustness) against the complexity of signal acquisition and processing.

According to the four-level classification scheme for non-invasive strategies (Lobo-Prat et al., 2014), the IDS adopted in this project is positioned as follows:

- **Level 1 (Local Source):** Vocal apparatus / User’s voice.
- **Level 2 (Physiological Phenomenon):** Speech production.
- **Level 3 (Measured Signals):** Acoustic signal (voice command).
- **Level 4 (Sensors):** Microphone / Audio acquisition system (on-board or external).

For a voice-command system, the number of potential states is theoretically large. However, practical limitations arise from software and computational resources and, in some cases, internet connectivity requirements for speech recognition. Moreover, accuracy may degrade in noisy environments or due to interfering voices, and the use of explicit keywords can be **disruptive** in certain social contexts (e.g., quiet environments or during conversation).

The system’s high-level control relies on the synergy between a Python script using the **Vosk** offline library [4] and the **Arduino Nano 33 IoT**.

4.4 Protocol for testing and validation

The validation strategy, which was carried out, uses the V-Model approach, dividing the validation into test of single components (Component Testing) and test of the entire system (System Testing). This was made to verify that the functional and safety requirements, defined in the previous chapters, are satisfied.

Component test protocol

The aim is to isolate and verify the performance of voice recognition and motion actuation.

1. **Test C1:** Intention detection accuracy (voice recognition)

- **Procedure:** in a controlled environment, with low noise level, it has been said a sequence of at least 10 valid vocal commands and invalid commands.
- **Outcome measures:** computation of a success rate (%) as the vocal commands correctly recognized over the total number of given commands.

2. **Test C2:** Trajectory tracking performance

- **Procedure:** trajectory tracking performance was evaluated on 5 healthy participants, for each participant, 5 repetitions of each predefined trajectory were recorded for all paths. Tracking error was computed as the point-wise difference between reference and measured joint angle over time.
- **Outcome measures:**
 - RMSE (Root Mean Square Error): RMSE between the reference trajectory (q_{ref}) and the trajectory measured by the encoder (q_{meas}) to look at precision.
 - Overshoot: verify that the final reached angle does not overtake the maximum angle.

System test protocol

The aim of these tests is to validate the complete interaction between user, computer and hardware.

1. **Test S1:** Full gameplay loop latency

- **Procedure:** this test simulates a complete game of the Vocal trivial pursuit. It is done to verify the complete sequence of action.
- **Outcome measures:**
 - System latency (T_{lat}): It is the time between the vocal command pronounced by the user and the beginning of the movement of the motor.
 - Task completion rate: it is the percentage of the game rounds successfully completed without communication error or software failure.

2. **Test S2:** Safety validation (Stuck condition strategy)

- **Procedure:** it simulated a physical block of the arm while it was performing a movement along the trajectory.
- **Outcome measures:**
 - Reaction time (T_{rea}): it is the time needed by the system to recognize the position error (difference between q_{ref} and q_{meas}) and disable the motor.
 - Safety outcome: pass or fail validation of the fast disabling of the motor to avoid user injury.

5. Results

This section reports the experimental results of the Vocal Trivial Pursuit system validation. System performance is evaluated against the functional and non-functional requirements defined in *Section 3*, following the testing protocol in *Section 4.4*. The analysis focuses on vocal command reliability, elbow kinematic accuracy, and safety mechanisms. Results are based on at least **10** repetitions per test and are reported as mean values.

Intention Detection and Communication Performance

To verify the **Voice command set** requirement, 30 vocal commands were tested. The system recognized the predefined game-related commands (Blue, Green, Yellow) with an accuracy of **90%** in a quiet environment using an external microphone and 76% without it. Unrecognized or incorrectly classified commands did not trigger any motion, confirming safe rejection of invalid inputs. The **Command-to-target mapping** requirement was verified by observing that each correctly recognized command consistently triggered the same predefined elbow target position, with no ambiguity across trials. Regarding **System latency** and **Safety on invalid input or disconnection**, vocal command recognition averaged **1.65 s** with a microphone and **2.05 s** without it. Including Wi-Fi transmission and controller response, the average time from command execution to motion onset was approximately $T_{lat} = 2.57$ s. Although this exceeds the defined latency threshold, system behaviour remained consistent and predictable. In case of recognition errors, missing commands, or wireless disconnection, no motion command was issued and the system remained in a non-actuated safe state, awaiting a new valid command.

Experimental Kinematic Validation

The elbow motion was constrained to a functional **range of motion** between 0° and 90°. To validate the kinematic behaviour of the system, a reference trajectory was established by recording at least ten repetitions of the movement in both no-load and loaded conditions (corresponding to the weight of the patient's arm). The recorded data were subsequently processed in MATLAB.

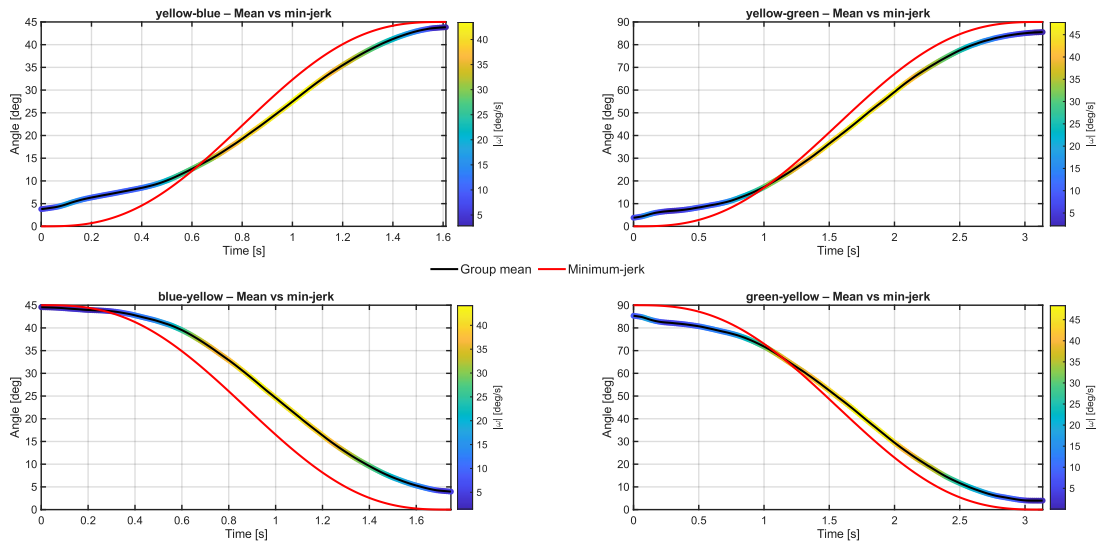


Figure 4: Comparison between the ideal MJ trajectory and the real one with velocity representation during the motion

The measured trajectories shown in the figures represent the average profile computed across multiple subjects, each performing a long-duration session using the same hardware configuration, in order to assess repeatability and robustness of the system behaviour. The experimental validation was conducted by comparing the averaged measured trajectories against the analytically defined minimum-jerk reference.

Fig. 4 shows the comparison between the minimum-jerk trajectory and the experimental motion, together with the corresponding velocity profiles. As shown in the figure, velocity is zero at the beginning and at the end of the movement, while the peak velocity occurs approximately at the midpoint of the motion, consistently with the expected kinematic behaviour.

Target Reaching Accuracy

The system's ability to reach discrete target positions (0° , 45° , 90°) was evaluated across all trials. The elbow consistently reached the commanded angles with an average error, computed with respect to the minimum-jerk polynomial, always **below 5°** at the final elbow position, satisfying the functional requirement for unambiguous answer selection during the Trivial Pursuit task. In Fig. 5, the tracking **error over time** is reported as the **mean error computed across all trials**. Additionally, the figure includes the **mean error over time for each subject and trajectory**, together with the associated **95% confidence interval**, in order to capture inter-subject variability throughout the movement.

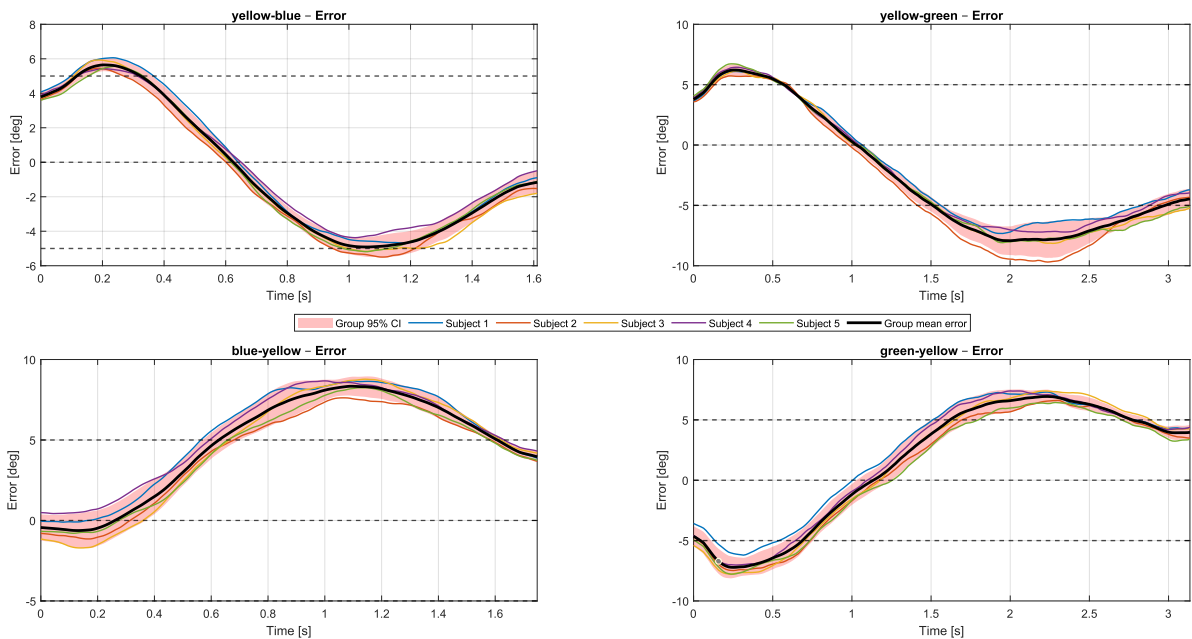


Figure 5: Tracking error over time averaged across trials and subjects, with 95% confidence interval.

From the tracking error plots, a **non-negligible discrepancy** between the analytical minimum-jerk reference and the experimentally measured trajectories clearly emerges. To further investigate the origin of this error, the system behaviour was analysed through the complete hysteresis loop shown in Fig. 6. The hysteresis loop highlights a systematic mismatch between the analytical minimum-jerk model and the trajectories executed by the hardware, indicating that the ideal kinematic model does not fully account for the exoskeleton's physical constraints.

For this reason, the tracking error was evaluated using two references: the analytical minimum-jerk trajectory and an empirical no-load reference. A **clear reduction** in tracking error is observed when comparing the measured trajectories against the no-load reference, suggesting that a significant portion of the apparent error with respect to the minimum-jerk model arises from intrinsic **hardware limitations** rather than control inaccuracies. The corresponding error distributions are reported as boxplots in Figs. 7 and 8, enabling a clearer distinction between model-related discrepancies and hardware-induced limitations.

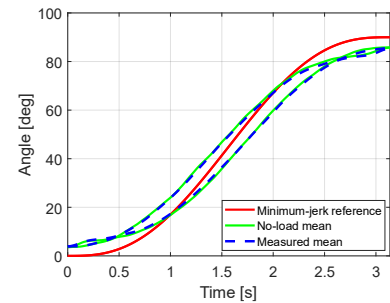


Figure 6: Hysteresis loop ($0^\circ \leftrightarrow 90^\circ$)

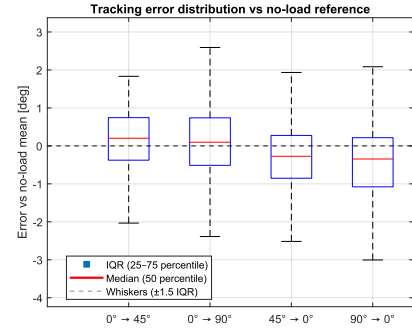
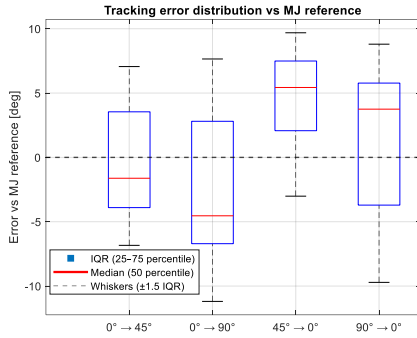


Figure 7-8: Tracking error boxplots with respect to the minimum-jerk reference (left) and the no-load reference (right).

In addition, the root mean square error (RMSE) distributions for each trajectory are reported in Figs. 9 and 10, providing a compact quantitative comparison of tracking performance with respect to both references.

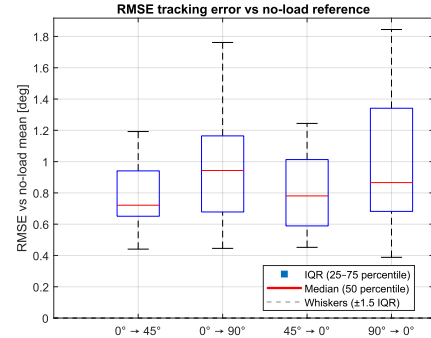
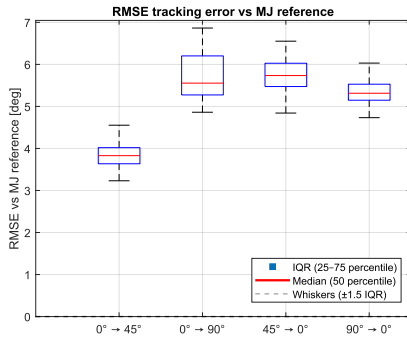


Figure 9-10: RMSE boxplots with respect to the minimum-jerk reference (left) and the no-load reference (right).

Usability

Subject	Joint alignment	Wearing comfort	Movement quality
1	6/10	8/10	8/10
2	6/10	4/10	6/10
3	8/10	8/10	9/10
4	8/10	6/10	8/10
5	4/10	6/10	6/10

Table 1: Usability scores assessed on healthy subjects

The usability evaluation in Tab. 1 highlights good perceived movement quality across **healthy subjects**, with consistently high scores. Wearing comfort shows moderate variability, indicating subject-dependent fit and contact perception. Joint alignment receives lower and more scattered scores, suggesting that anatomical alignment is the main usability limitation of the current setup in **non-impaired users**.

Admittance Control and Safety Validation

With respect to safety control, the system was originally intended to include an admittance-based strategy relying on the embedded force sensor to detect abnormal interaction forces. Due to hardware limitations, a **stuck-condition safety mechanism based on position feedback** was implemented instead. If the elbow remained stationary for 10 consecutive samples at 20 Hz while a motion command was active, the Arduino detached the servomotor, corresponding to a reaction time $T_{rea} = 0.5 \text{ s}$ preventing potential injury. This mechanism allowed to verify of both the **Stuck condition safety** and **Sensor-based position feedback** requirements: across a multi-subject validation, **20 repeated test trials per subject** consistently detected the stationary condition using joint position measurements and reliably triggered motor detachment.

6. Discussion and conclusion

The development of the Vocal Trivial Pursuit project demonstrates the feasibility of combining low-cost mechatronic hardware with high-level software logic to provide safe and fully assisted elbow motion for subjects with severe upper-limb impairments. By integrating a voice-based cognitive trigger implemented in Python with low-level motion control executed on an Arduino platform, the system enables **intuitive interaction** while maintaining predictable and **supervised behaviour** throughout task execution. The proposed solution confirms that cognitive intent can be effectively translated into reliable motor assistance in a constrained and safety-critical scenario.

A key technical insight emerging from this work is the relevance of **minimum-jerk trajectories** in assistive human-robot interaction. While constant-velocity profiles are adequate for many industrial applications, minimizing jerk is crucial in assistive contexts to enhance comfort and reduce impulsive transients that may negatively affect fragile users. Experimental observations confirm that minimum-jerk motion generation leads to **smooth, repeatable, and predictable trajectories** that are well suited to safety-oriented position control.

During the development phase, an admittance-based compliance strategy was initially considered to further improve interaction safety through force-based adaptation. However, due to hardware limitations affecting the reliability of force sensing, the control architecture was redesigned to rely on predefined minimum-jerk trajectories combined with a **stuck-condition safety algorithm** based on position feedback. This solution proved effective in detecting abnormal motion conditions and promptly disabling actuation in the presence of unexpected obstructions, ensuring user protection despite the absence of active force-based compliance.

From a system-level perspective, the **implemented solution satisfies the requirements** of the Vocal Trivial Pursuit task, enabling reliable command execution and unambiguous target selection. Some trajectory-dependent inaccuracies were observed for specific target transitions, highlighting the influence of **hardware limitations** and interaction dynamics. These effects do not compromise task correctness and can be mitigated at the task-design level, for example by adjusting the spatial separation of selectable targets to preserve robustness.

Despite the successful validation of the motorized elbow joint, the shoulder joint remains stabilized through a passive torsional spring, limiting the overall workspace. Moreover, performance degrades in **noisy environments**, highlighting the limitations of voice-based control. Future developments will therefore focus on motorizing the shoulder to enable fully assisted **multi-degree-of-freedom** control and on exploring alternative intention detection strategies (such as **tongue movement**) or **noise-robust models** to accommodate users with different residual capabilities. In addition, integrating a **stereoscopic vision system** could enable real-time object tracking through blob analysis, allowing trajectory reconstruction and quantitative assessment of interaction dynamics. This information could be exploited to adapt task difficulty and support therapeutic applications.

In conclusion, this prototype provides a solid foundation for more advanced assistive robotic systems, demonstrating that voice-based cognitive triggers can enable motor-impaired individuals while maintaining **safety**, predictability, and **user comfort**.

7. References

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